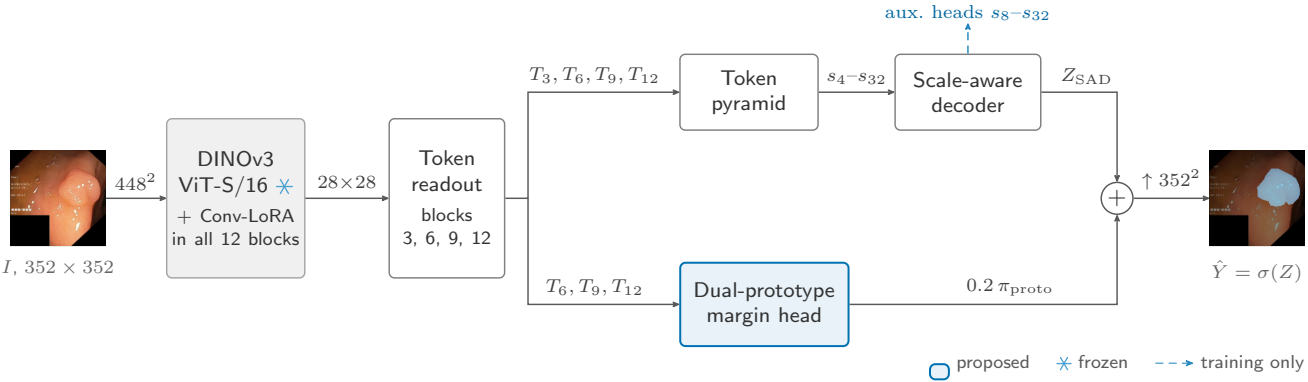

aux. heads s_8–s_{32}
I, 352×352
448^2
DINOv3
ViT-S/16 ✳
+ Conv-LoRA
in all 12 blocks
28×28
Token
readout
blocks
3, 6, 9, 12
T_3, T_6, T_9, T_{12}
Token
pyramid
s_4–s_{32}
Scale-aware
decoder
Z_{SAD}
$\uparrow 352^2$
$\hat{Y} = \sigma(Z)$
T_6, T_9, T_{12}
Dual-prototype
margin head
$0.2\,\pi_{\mathrm{proto}}$
proposed
✳ frozen
training only